Q1) Convert D-FF into divide by 2.

What is the max clock frequency the circuit can handle, given the following information?

T_setup= 6nS

T_hold = 2nS

T_propagation = 10nS

Ans1)q bar has to be connected to the input of d-flip flop.

So the max clock frequency would be 1/(16 ns)

(max freq = progation delay + set up time)[/quote]

Question(2)

Why do we gradually increase the size of inverters in buffer design when trying to drive a high capacitive load? Why not give the output of a circuit to one large inverter

Ans2)

Reduction of energy dissipation in CMOS digital circuits has become an important goal of the design optimization process. Optimization tools rely on accurate and efficient energy analysis and estimation techniques. These techniques in turn need to account for all of the key components of the energy consumption in CMOS circuits. One such component, which is generally referred to as the short circuit or rush-through energy dissipation, is the energy consumed by flow of current from Vdd to Gnd through a direct current path that is temporarily

established during an output transition. Short circuit energy dissipation is becoming an important factor as the number of buffers increase. Without considering the short circuit energydissipation, sizing a multi-stage buffer to drive a large capacitive load may result in a poor solution in terms of the energy-delay product.

Ans(2)

We cannot use a big inverter to drive a large output capacitance because, who will drive the big inverter? The signal that has to drive the output cap will now see a larger gate capacitance of the BIG inverter.So this results in slow raise or fall times .A unit inverter can drive approximately an inverter thats 4 times bigger in size. So say we need to drive a cap of 64 unit inverter then we try to keep the sizing like say 1,4,16,64 so that each inverter sees a same ratio of output to input cap. This is the prime reason behind going for progressive sizing.

Question(3)

Why don’t we use just one NMOS or PMOS in a transmission gate?

ANSWER : Q3

PMOS degrades Logic 0 & NMOS degrades logic 1

To restore the logic levels to full, both NMOS & pMOS will be used together in TG

Ans(3)

Using only an nmos will result in an poor 1. Why is it so? Assume the gate voltage on NMOS is 5V. If we connect Drain to 5V, and the source is initially at 0, NMOS will turn on as long as there Vgs >Vth, this means, once the source reaches 4.3V (Assuming Vth=0.7), the nmos will turn off and there will be no more increase in source voltage.Similarly the opposite happens with PMOS, it doesn't give us a clean 0, but it can give a full 5V. So we use a combination of both NMOS and PMOS so that our signal doesn't get degraded by Vth on

either side of VDD and GND.

Question(4)

Is there in Hold violation in the Circuit of Q1?

Describe clearly when there will be Hold violation wrt to the given data and how we can solve it in circuit level?

Answer(4)

There is no hold vioaltion in the given circuit. As the hold time is less than the propagation delay.

Otherwise buffers(even number of inverters) will be used in the feedback path in order delay the feedback reaching back the input.

Q(5)

In CMOS technology, in digital design, why do we design the size of pmos to be higher than the nmos.What determines the size of pmos wrt nmos. Though this is a simple question try to list all the reasons possible..

Ans(5)

In PMOS the carriers are holes whose mobility is less[aprrox half] than the electrons, the carriers in NMOS. That means PMOS is slower than an NMOS. In CMOS technology, nmos helps in pulling down the output to ground ann PMOS helps in pulling up the output to Vdd. If the sizes of PMOS and NMOS are the same, then PMOS takes long time to charge up the output node. If we have a larger PMOS than there will be more carriers to charge the node quikly and overcome the slow nature of PMOS . Basically we do all this to get equal rise and fall times for the output node.

Q(6)

How many unique boolean fucntions can be there for **n** number of inputs?

Ans

For n number of inputs, the possible number of min terms or max terms, k = 2^n

To form any bollean function, we can take any combination of these.

So possible boolean functions = kc0 + kc1 + kc2 ...+kck = 2^k = 2^2^n

Q7

Design a sequential circuit which cuts the every second pulse from the input(clk)?

Hint: If we think in other way, it is nothing but frequency devider by 2 , But with 25% Duty cycle!!!

A7)

First do a Divide by 2 counter, ie connect Qbar to D input of FF.

Connect the Q output and CLK to a 2 input AND gate, this will gives us a divide by 2 clock with 25% duty cycle.

Question8

Guys this is the basic question asked most freqently. Design all the basic gates(NOT,AND,OR,NAND,NOR,XOR,XNOR) using 2:1 Multiplexer.

Ans8

Using 2:1 Mux, (2 inputs, 1 output and a select line)

(a) NOT

Give the input at the select line and connect I0 to 1 & I1 to 0. So if A is 1, we will get I1 that is 0 at the O/P.

(b) AND

Give input A at the select line and 0 to I0 and B to I1. O/p is A & B

(c) OR

Give input A at the select line and 1 to I1 and B to I0. O/p will be A | B

(d) NAND

AND + NOT implementations together

(e) NOR

OR + NOT implementations together

(f) XOR

A at the select line B at I0 and ~B at I1. ~B can be obtained from (a)

(g) XNOR

A at the select line B at I1 and ~B at I0

Q9

N number of XNOR gates are connected in series such that the N inputs (A0,A1,A2.....AN-1) are given in the following way:

A0 & A1 to first XNOR gate and

A2 & O/P of First XNOR to second XNOR gate

and so on..... Nth XNOR gates output is final output. How does this circuit work? Explain in detail?

Ans9

If N=Odd, the circuit acts as **even parity detector**, ie the output will 1 if there are even number of 1's in the N input...This could also be called as **odd parity generator** since with this additional 1 as output the total number of 1's will be ODD

If N=Even, just the opposite, it will be Odd parity detector or Even Parity Generator...

Q10)

All of us know how an inverter works. What happens when the PMOS and NMOS are interchanged with one another in an inverter?

Q11)

A good question on Layouts...

Give 5 important Design techniques you would follow when doing a Layout for Digital Circuits

Q12)

What is Latch Up? How do you avoid Latch Up?

Ans12

Coming to latchup problem,

If you look at the cross cectional view of any inverter, there is a positive feedback between a NPN and PNP transistor which results in latch up problem.This positive feedback results in excessive current which ultimately destroys the device. These NPN and PNP transistors are formed from the p+/n+ source/drains, p/n

well and substrates

Q13)

Implement the following circuits:

(a) 3 input NAND gate using min no of 2 input NAND Gates

(b) 3 input NOR gate using min no of 2 inpur NOR Gates

(c) 3 input XNOR gate using min no of 2 inpur XNOR Gates

Ans13

3 input NAND:

Connect :

i) A and B to the first NAND gate

ii) Output of first Nand gate is given to the two inputs of the second NAND gate (this basically realises the inverter functionality)

iii) Output of second NAND gate is given to the input of the third NAND gate, whose other input is C

((A NAND B) NAND (A NAND B)) NAND C

Thus, can be implemented using '3' 2-input NAND gates. I guess this is the minimum number of gates that need to be used.

3 input NOR:

Same as above just interchange NAND with NOR

((A NOR B) NOR (A NOR B)) NOR C

3 input XNOR:

Same as above except the inputs for the second XNOR gate, Output of the first XNOR gate is one of the inputs and connect the second input to ground or logical '0'

((A XNOR B) XNOR 0)) XNOR C

Q14:

An assembly line has 3 fail safe sensors and one emergency shutdown switch.

The line should keep moving unless any of the following conditions arise:

(i) If the emergency switch is pressed

(ii) If the senor1 and sensor2 are activated at the same time.

(iii) If sensor 2 and sensor3 are activated at the same time.

(iv) If all the sensors are activated at the same time

Suppose a combinational circuit for above case is to be implemented only with NAND Gates. How many minimum number of 2 input NAND gates are required?

ANS:

No of 2-input NAND Gates required = 6

You can try the whole implementation also.

Q15:

What is race-around condition? Explain in it in case of R-S Latch and solution to avoid that?

Ans:

The race around conidtion is : O/P osciallting between 0s & 1s.

This problem will occur in Latches especially if the clock is high for long time.

I can explain in case of J-K Latch
Suppose J=K=1 & O/P = Compliment of prev state (Q(t+1) = Q(t)')
So as far as clock is high, O/P oscialltes between 0 & 1

To avoid this,
they use Master-Slave configuration.

Q16:
What is metastability? When/why it will occur? Different ways to avoid this?
Ans : Q16

Metastable state: A un-known state in between the two logical known states.
This will happen if the O/P cap is not allowd to charge/discharge fully to the required loigcal levels.
One of the cases is: If there is a setup time violation, metastability will occur,
To avoid this, a series of FFs is used (normmly 2 or 3) which will remove the intermediate states.
Q18:
Give the state machine for a serial two's complimenter? Then design the complete circuit using DFF?
Hint: If you observe a binary number and its 2's compliment, the 0s will be retained until the first 1 occurs (from LSB side) and the first 1 also will be retained after that compliment all the follwoing bits.
It is very interseting & simple Qs to check the knowledge of state M/Cs

Q18:

The state M/C will have only two states. State A & B. Stay in State A as far as you are getting 0's and O/P is also 0. If 1 comes go to state B and O/P is 1.
If you are in state B, whether I/P is 1 or 0 stay in B only and O/P is compliment of input.

State Table is as follows:
PS x NS O/P
0 0 0 0
0 1 1 1
1 0 1 1
1 1 1 0

To obtain the circuit using DFF,
OR x & Q of FF and give at the I/P of FF
XORing of Q&x will give O/P(2's compliment)

Q19:
In a PLL, what elements(digital blocks) can be used to design the phase detector?
Ans: 19

1. XOR Gate
2. S-R Latch
3. PFD(Phase/freq detector) : It is designed from FFs & some NAND Gate COnnected to resetes)

Q20:

Describe a finite state machine that will detect three consecutive coin tosses (of one coin) that results in heads.

Assume state A : no heads has occured

state B: only one head has occured

state C: more than 2 heads has occured

So intial state is A

PS I/P NS O/P

A Tail A 0

A Head B 0

B Tail A 0

B Head C 0

C Tail A 0

C Head C 1

Q21:[/b]

What is Moore model & Mealy model? Explain.

Ans: Q21

A state machine consists of set of states, initial state, input symbols and transition function that maps input symbols and current state to next state.

Mealy machine: machines having outputs associated with transition of input and the current state. So in this case, the O/P will vary as soon as the input varies..O/P can't be held until clock pulses.

Moore machine: machines having outputs associated with only states. The O/P will vary only after the states are varied.So the cahgnes in the O/P will be only during clock transitions.

Adv & Disadv:

In Mealy as the output variable is a function both input and state, changes of state of the state variables will be delayed with respect to changes of signal level in the input variables, there are possibilities of glitches appearing in the output variables.

Moore overcomes glitches as output dependent on only states and not the input signal level.

All of the concepts can be applied to Moore-model state machines because any Moore state machine can be implemented as a Mealy state machine, although the converse is not true.

Q22)

How many minimum number of MOS transistors are required to implement a Full Adder using CMOS technology?

S = a xor b xor c

Cout = (a xor b) c + ab

Using these equations, we can implement,,,exact count of MOS transsitors is straight forward..

Note XOR can be implemented with only 4 2 input NAND Gates...

Q23:

(a) Show all the possible ways to convert a 2-input NAND Gate into an inverter?

(b) Show the implementaion of XOR Gate using minimum number of 2-input NAND Gates?

Ans: Q23:

(a) A 2 input NAND gate can be converted into an inverter in two ways: one way is by tieing up the two inputs and give the input, second give make one of the two inputs permanatly high and give the input at the other input.

(b) A XOR B = AB' + A'B = (AB)' A + (AB)' B

So one 2-input NAND is needed to generate (AB)' 3 other to implent the rest of boolean function. So total we need 4 2input NAND Gates.

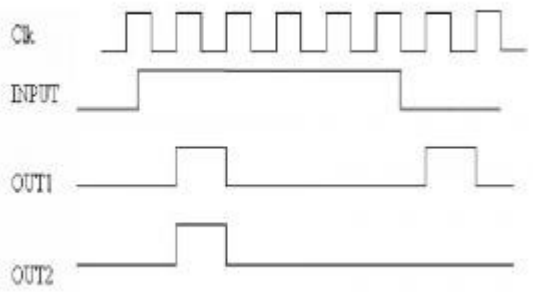
Q24:

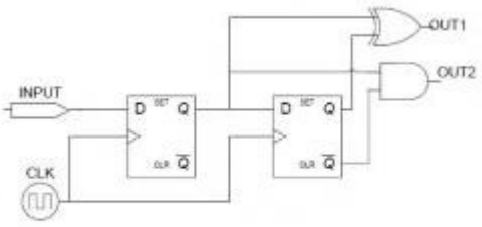
It is required to connect a Master, which generates data @ 200 Mega Samples/sec to a Slave which can receive the data @ 10 Mega Samples/Sec. If the data **burst** in 10Micro Sec, what is the optimal size of FIFO to be used to avoid lose of data?

Ans : (200-10) * 10 = 1900 samples is the size of FIFO

Q25:

The clock and the input output wavefroms are shown in the follwoing URL:





Q27:

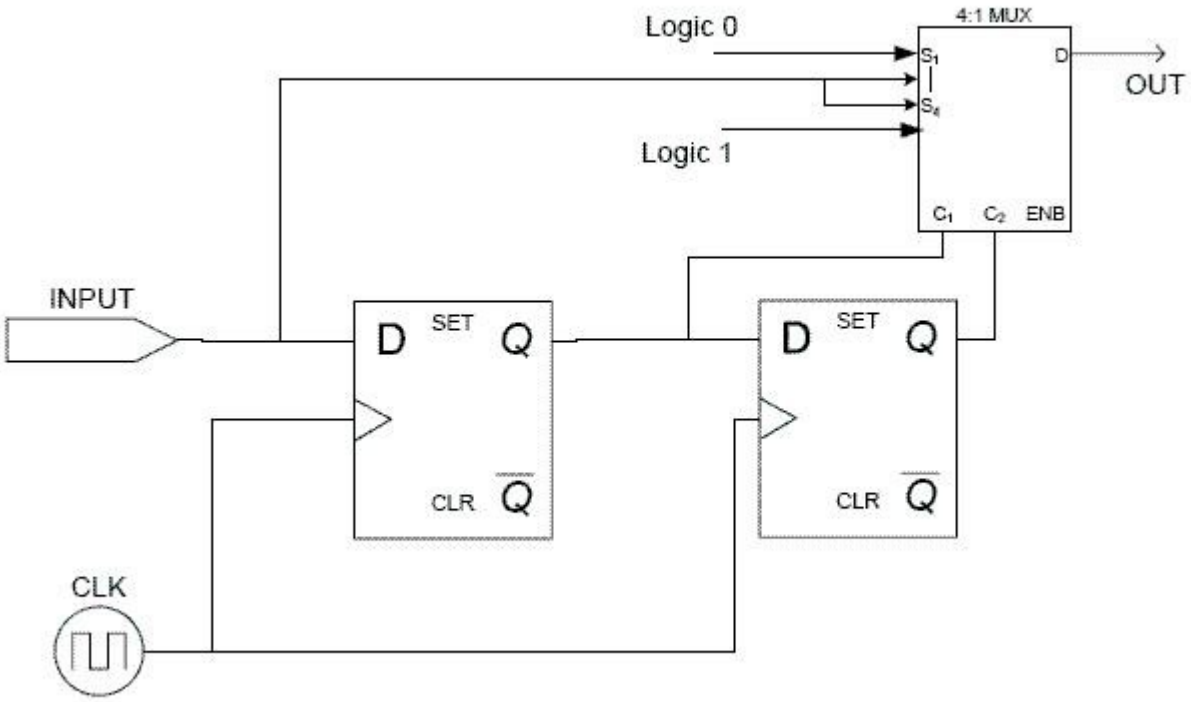
Design a synchronous sequential circuit to check the highest number of ones and zeros in the last 3 input samples. Your ckt should give 1 at the O/P if the last 3 samples at the input has more 1's similarly 0 when the no. of zeros is high.

Eg:

IN : 001110110000
OUT: 0111111000

Constraints:

1) You are supposed to use only Multiplexers and DFFs for your design. No external gates. To be specific, 1 4:1 Mux only.
2) Design should be optimized one.
3) Only one clock is available to you. And it is given that the input is sampled at that clock rate only.



The above diagram shows the design of the circuit for Q27, using 2 DFFs and one 4:1 Mux as specified in the design constraints. The 4 inputs are named from S1-S4 and 2 seclect lines are shown as C1 & C2 where as D is the O/P of the MUX.

Q28 What is overflow? How can you detect overflow in signed and unsigned numbers?

Ans: Q28:

case1 : Unsigned numbers:

In N-bits, we can represent numbers from 0 to (2^N) - 1. Suppose if we are adding 2 N bit unsigned numbers and if the result is greater than (2^N) - 1 , overflow will occur. To detect this, check whether the

MSB addition (Nth bit) + Carry generated from N-1bit addition is genrating any carry or not. If there is carry out, there is overflow.

case2 : Signed numbers:

In N-bits, we can represent numbers from -(2^(N-1)) to (2^(N-1)) - 1. Suppose if we are adding 2 N bit signed numbers and if the result is not in the above range , overflow will occur. To detect overflow in this case : if two numbers with the same sign (either positive or negative) are added and the result has the opposite sign, an overflow has occurred.

Ans: Q28:

Forgot two more cases in case of signed...

So overall, the conidtions to detect overflow are here again:

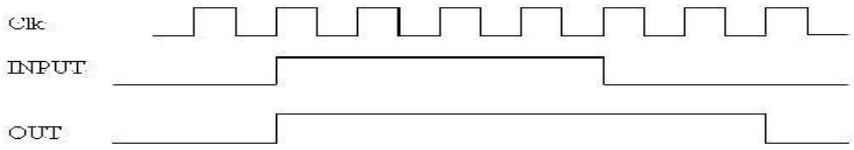
In unsigned arithmetic a carry out of the most significant digit means that there has been an overflow

A signed overflow has occurred in an addition or subtraction if:

* the sum of two positive numbers is negative;
* the sum of two negative numbers is positive;
* subtracting a positive number from a negative one yields a positive result; or
* subtracting a negative number from a positive one yields a negative result.

Q29:

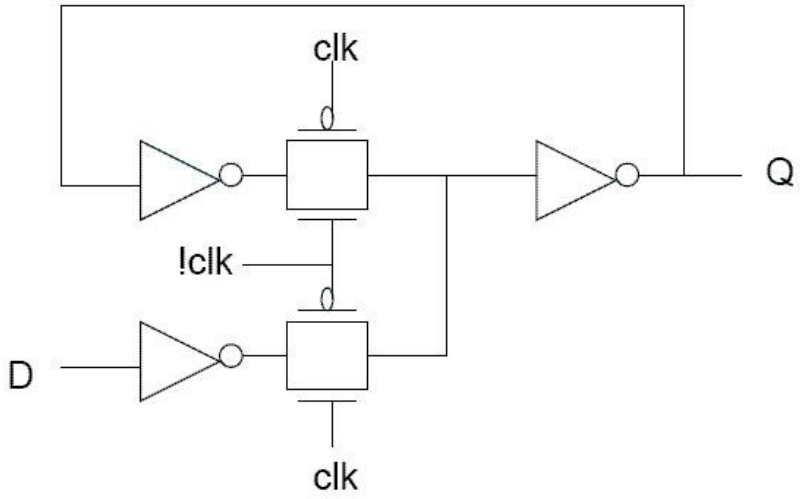
Give the circuit to extend the falling edge of the input by 2 clock pulses. The waveforms are shown in the following figure.

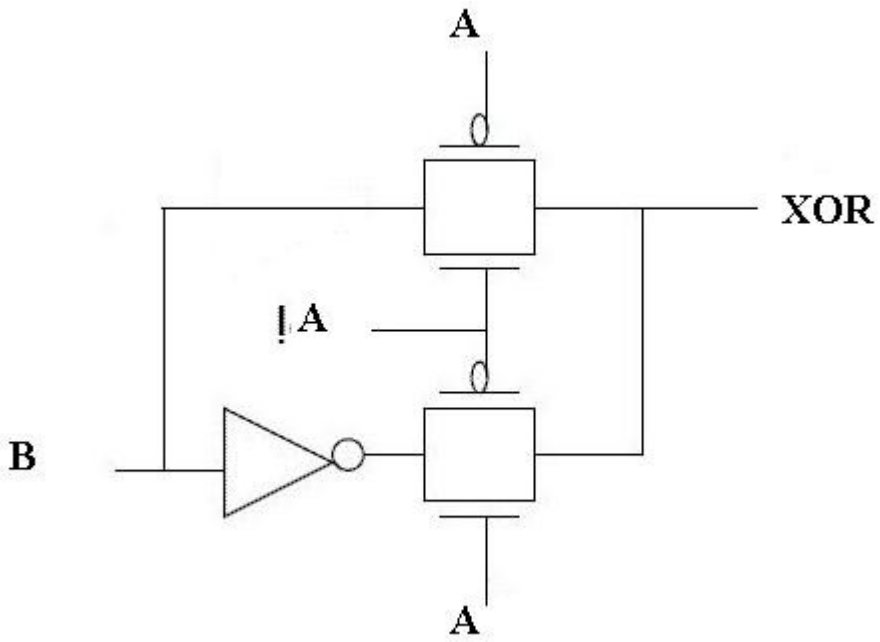


Q30)

(a) Draw a Transmission Gate-based D-Latch.

(b) Design a Transmission Gate based XOR. Now, how do you convert it to XNOR? (Without inverting the output)





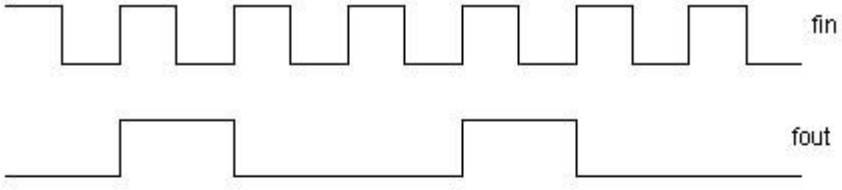
Q31,Q32,Q33:

Design the Digital Circuit which gives

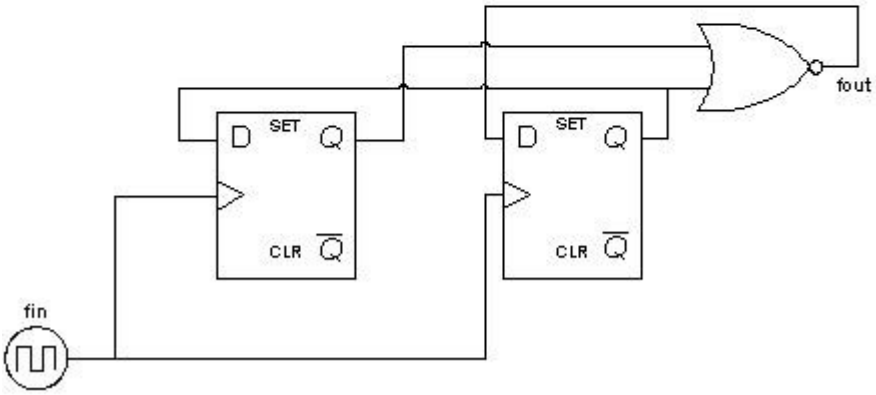
(Q 31) fout = (1/2) fin

(Q 32) fout = (1/3) fin

Waveforms:



Design:



The above circuit gives a duty cycle of 1/3rd (that is 33.333)

To get 2/3rd(that is 66.667) , use NAND gate instead of NOR gate

(Q 33) fout = (2/3) fin (3 different circuits)

NOTE:

(a) fout is O/P freq and fin is I/P freq

(b) Duty cycles are also not mentioned..so its okay to design for any duty cycle.

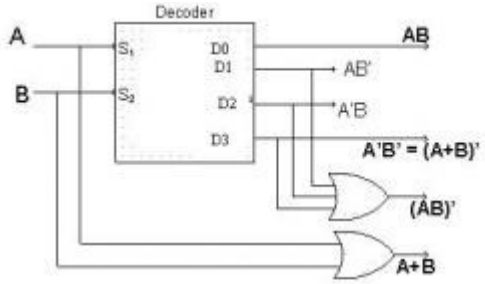
(c) All the ckts design using DFFs and min no of external gates

Q34:

You are given a 2:4 decoder. 2 input OR gate and one 3 input OR gate. Using these Components design the following system whcih takes A & B as inputs and generates the 4 O/Ps : AB, (AB)' , A+B, (A+B)' .

Ans: Q34:

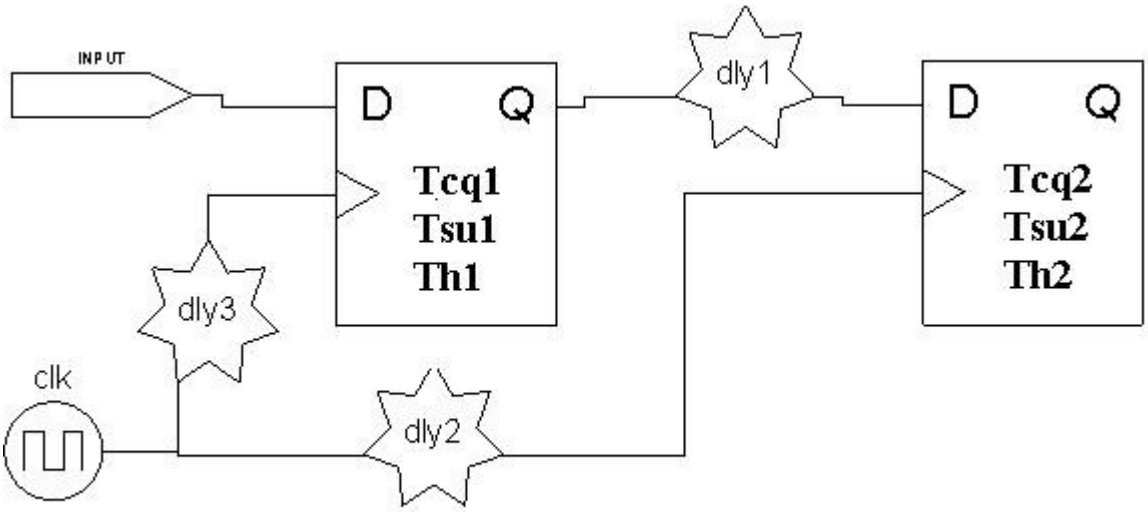
2:4 decoder will have 4 O/Ps which are the minterm/maxterms of the 2 inputs. So the O/P are AB, AB', A'B, A'B'. So AB and (A+B)' = A'B' are directly the O/P s of decoder. Whereas A+B can be obtained using 2 input OR gate(which is given). So only O/P that is needed is (AB)' = A' + B' = A'(B+B') + B'(A+A') = AB' + A'B + A'B'. That is , use 3-input OR gate for this. The whole design is shown below.



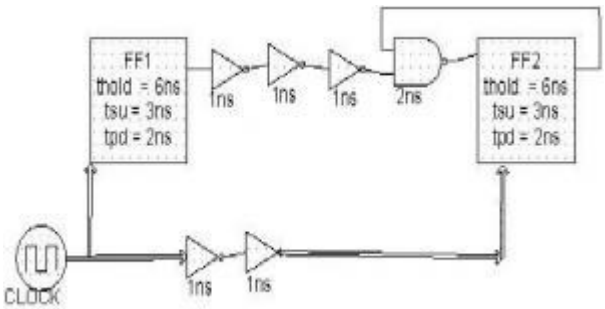
Q35:

The following digital circuit shows two flops with a logic delay (dly1) in between and two clock buffer delays (dly2, dly3). Derive the conditions in terms of (dly1,dly2,dly3) to fix setup and hold timing violations at the input of second FF?

Tcq -- Clock to Q delay, Tsu -- Setup time and Th -- hold time.



Q36:



(a) For the Circuit Shown above, What is the Maximum Frequency of Operation?

(b) Are there any hold time violations for FF2? If yes, how do you modify the circuit to avoid them?

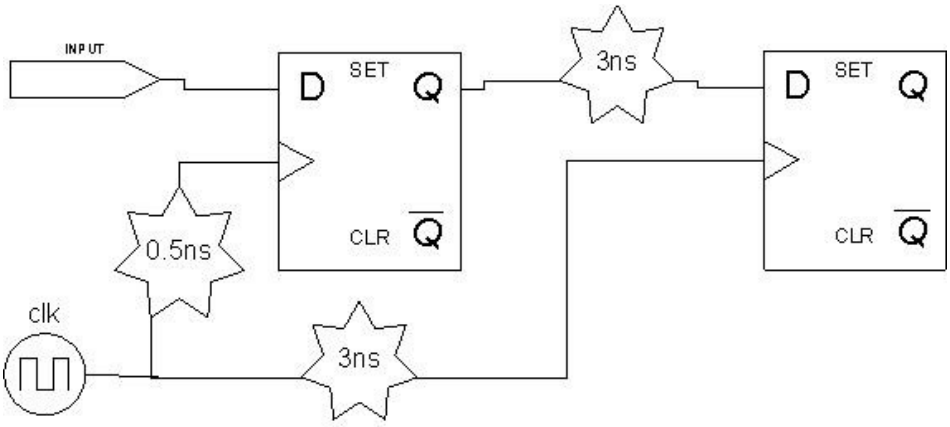
Ans36

The minumum time period = 3+2+(1+1+1) = 8ns

Maximum Frequency = 1/8n= 125MHz

There is a hold violation in th ecircuit.You can avoid it by giving the input to the AND gate through two inverters.

Q37:



If both the flip flops have same clock-to-Q delay of 2.5ns,setup time of 2ns and a hold time violation of 1ns, what is the maximum frequency of operation for the circuit shown above?

ANS:

Considering the set up time violation,

T>= Tc-q2 + Tsu2 + dly - Tskew

As Tskew = 3-0.5 =2.5ns

T>= 2.5 +2+ dly -2.5 ns

T>= dly +2 ns

Considering the hold time violation,

Thold +Tskew <= dly + Tc-q

1+2 <= dly + 2.5

From this dly >= 0.5ns in order to meet hold constraints

To obtain maximum frequency subsitute the value of dly = 0.5ns in the equation 1 gives

T>= 0.5 + 2 ns

which means the maximum frequency is (1/(2.5n)) = 400 MHz.

Ans:Q42:

Setup time :

(T/2) + delta >= Tcq1 + dly1 + Tsu2

Hold time:

Th2 <= delta + Tcq1 + dly1

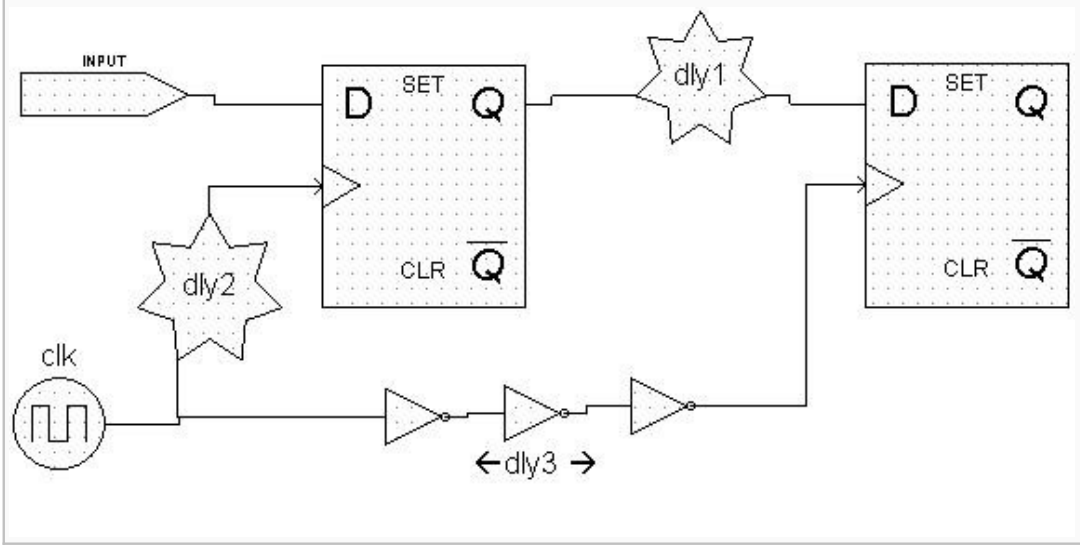
where delta = dly3 - dly2, assuming positive skew
and T is clock period.

Note: The procedure is same as that of Q37.Just draw the waveforms with proper delays, you will get above equations.

srikanth wrote:

Q42:

Derive setup time/hold time vilation equations for the follwoing circuit? Assume Tcq1 Clock to Q delay, Tsu1 -- Setup time and Th1 hold time for first FF and similarly Tcq2,Tsu2,Th2 for second FF.

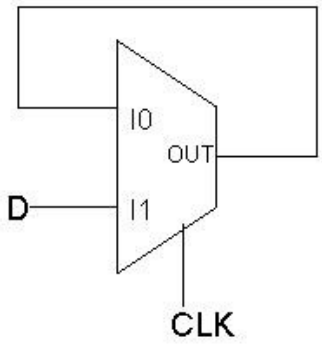


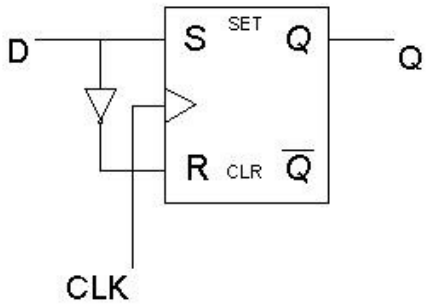
Q43:

Design a D-latch using
(a) using 2:1 Mux
(b) from S-R Latch

Ans:43:

(a) D-latch using 2:1 Mux
(b) D-Latch from S-R Latch





Q44:

Suppose A & B are two unsigned n-bit numbers, how many minimum number of bits required for Y = A + B + (A*B). Here + is for addition and * is for multiplication. All are unsigned operations only.

Ans:44)

In n bits, the maximum number is 2^n - 1.

So maximum value for Y is

(2^n - 1) + (2^n - 1) + (2^n - 1) (2^n - 1) = 2^2n - 1

So the number of bits required for Y are 2n

Q45:

(a) Give the truth table of a Half Adder?

(b) Design a full adder from HA's? (You can use Min no. of external gates)

Ans:Q45:

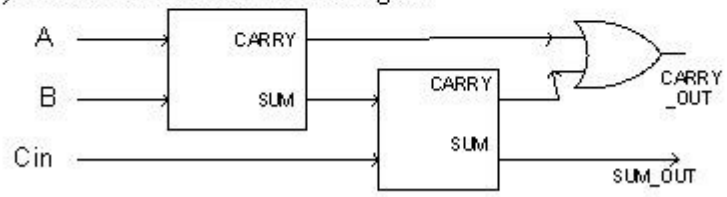
If we observe that clearly...sum will be sum of the two half adders

and carry will be 1 if any of the HAs or both HA has a carry...thats why that OR Gate

(a) Truth table of a HA:

A	B	Cout	Sum
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

(b) Full adder from 2 HA and one OR gate:



Q46:

(a) How will you count the number of 1's that are present in a given 3-bit input using full adder?
(b) If input is a 7-bit vector, how many minimum number of full adders are required to count the number of 1s?

Q47) Came across this question from a friend.
Design a circuit that calculates the square of a number? It should not use any multiplier circuits. It should use Multiplexers and other logic.
A47)This is interesting....
1^2=0+1=1
2^2=1+3=4
3^2=4+5=9
4^2=9+7=16
5^2=16+9=25
and so on
See a pattern yet?To get the next square, all you have to do is add the next odd number to the previous square that you found.See how 1,3,5,7 and finally 9 are added.Wouldn't this be a possible solution to your question since it only will use a counter,multiplexer and a couple of adders?It seems it would take n clock cycles to calculate square of n.
Let me know if my thinking is correct.

A 48

DFT means design for testability.
Other than the functional logic,you need to add some DFT logic in your design.
This will help you in testing the chip for manufacturing defects after it come from fab.
Scan,MBIST,LBIST,IDDQ testing etc are all part of this.

Q54) In CMOS deisgn, given a choice between implementing a logic in NOR and NAND implementation,

which one would you prefer and why?

Ans 54. If it is a complementary CMOS Nand is preffered over NOR as NOR has PMOS in series which slows it down.

If it is a pseudo-NMOS, NOR is preferred as it has Transistors in parallel.

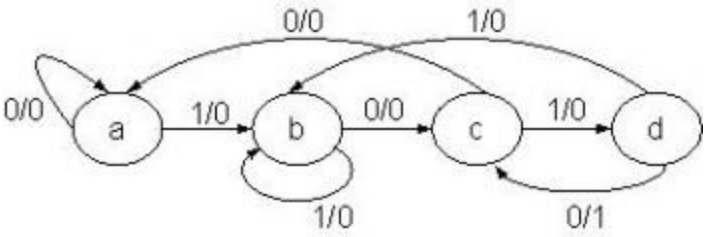
Q55) What are the conditions for obtaining worst case set up and hold times?

Q57:

Give the State Machine for detecting the sequence â€œ1010â€ö from a serially coming data for both (a) Overlapping & (b) Non-overlapping cases.

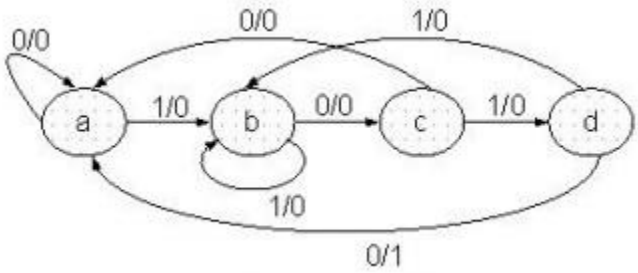
Ans Q57:

Part (a) Overlapping case



Ans Q57:

Part (b) Non-overlapping case



Q58:

Sender sends data at the rate of 80 words / 100 clocks Receiver can consume at the rate of 8 words / 10 clocks Calculate the depth of FIFO so that no data is dropped. Assumptions:

There is no feedback or handshake mechanism.

Occurrence of data in that time period is guaranteed but exact place in those clock cycles is indeterminate.

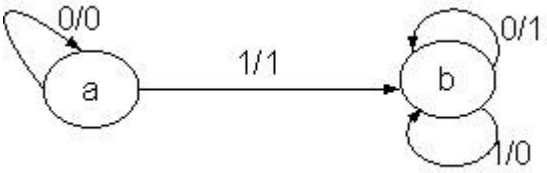
A58) : In the worst case,sender would send 80 words in 80 clock cycles.In this time period,receiver would only be able to receive 8*8=64 words in those 80 clock cycles.Therefore FIFO size should be 16 words.

I may well be mistaken but I will fall back on the old adage that it is only through making mistakes that you will learn something! ANS: 32 words (back to back 20/80/80/20)

Q60:

One more intersting Q on FSM

What does the following FSM do?

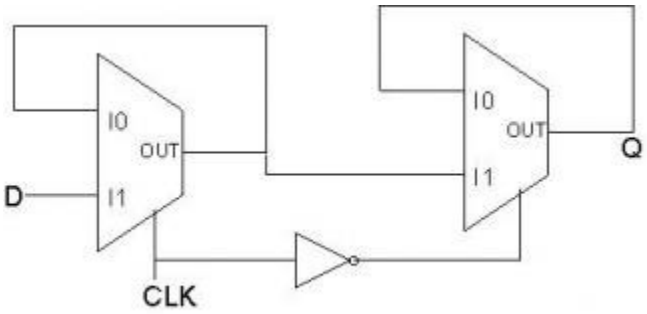


If you observe the state machine, it shows that the system retains 0 as it is and the first 1. After one 1 has come, it compliments all the following bits. One more thing...The data starts with LSB

Q66)

How will you implement a Master Slave flip flop using a 2 to 1 mux?

Ans: Q66:



Q67:

Using DFFs and minimum no. of 2Ã—1 MUXs, implement the following XYZ flip-flop.

X Y Z Qnext

0 0 0 1

0 0 1 0

0 1 0 0

0 1 1 1

1 0 0 Q

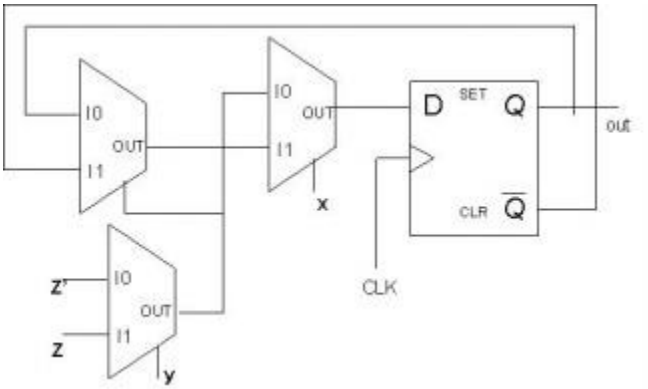
1 0 1 Q`

1 1 0 Q`

1 1 1 Q

Ans:Q67:

I could come with a solution of 3 2:! Mux and 1 DFF
If compliments are not availble, we need one more 2:! Mux for finding z'



ANS 67

The above answer was for a n-bit adder.. which has
the following inputs..
1. n-bit subtraend
2. n-bit minuend
3. one bit for carry input.

it has the following outputs
1. n-bit sum.
2. one carryout bit.
So the above was a implementation for a ripple binary adder.

As you said,the logic equation for the sum for both adder and subtractor is A xor B xor Cin.
carry out for both is different though.
Cout for full adder is
Cout = AB + BC +AC.

Cout for full subtractor is..
Cout= A_bar B + A_bar C + BC.

so full subtarctor can be implemented by inserting a not gateand inverting A.

Divide it into 2 parts.
For the first four..
A 2*1 mux will have a enable signal as x.
If x= 0, then Io= 1 wil be used to activate the 4* 1 mux.
4*1 mux wil have 2 select signals y and z. and 4 inputs. being

Ino= 1, in1=0,in2=0 in3=1.
The output of the 4* 1 mux is sent to a D flip flop.
The lower four combinations of x,y,z can be implemented,
when the select signal of the 2* 1 mux x=1 then I1= Q is selected. which wil be sent as the input to the second D-flip flop.
The Qnext of both the flip flops are sent to a 2*1 mux. whose enable signal is x.
the final output Q_next wil be sent as a input to the I1 pin of the 2*1 mux.
it can be implemented using 3 mux and two flip flops.
I am not sure if you caould understand anything out of it.
But i put it in words because i was not ablue to cut and paste in the post

the lower parts can be imlemented using a single D flip-flop as they dont depend on the y and z values.

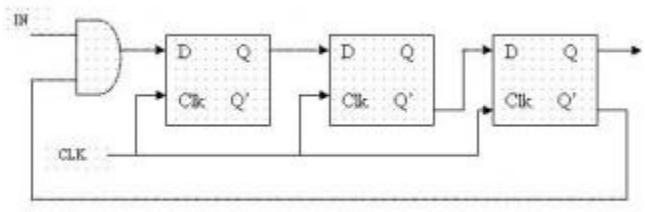
Q69:
In what cases do you need to double clock a signal before presenting it to a synchronous state machine?
A 69
When the signal is asynchronous (Probably coming from a different clock domain)

Q 69) How will you implement a Full subractor from a Full adder?
Ans69)
A subtractor should subtract subtraend from minuend. Subtraction is nothing but addition of the two's compliment of the subtraend.
2s compliment is negation of the number and adding 1 to it.
1st compliment can be implimented by xoring the subtraend with 1.
num Xor 1 = compliment of num.
Full subtractor can be implemented using a full adder in the following manner.
all the bits of subtraend should be connected to the xor gate. Other input to the xor being one.
The input carry bit to the full adder should be made 1.
Then the full adder works like a full subtractor.

Q71) A very good interview question...
What is difference between setup and hold time. The interviewer was looking for one specific reason , and its really a good answer too..The hint is hold time doesn't depend on clock, why is it so...
ANS: Setup violations are related to two edges of clock, i mean you can vary the clock frequency to correct setup violation. But for hold time, you are only concerned with one edge and does not basically depend on clock frequency.
Q72:
You have three delay elements D1, D2, D3 that delay a clock by 25%,50% and 75% respectively. Design a frequency doubling (fout = 2 * fin) circuit that uses these delay elements along with any combinational logic.
ANS: I think we can double the frequency by just using the delay element D1 and an XOR gate. Just pass the input signal through D1. Now the delayed signal and the original signal are input to a 2-i/p XOR.

Q73:

What are all the test patterns needed at the input, IN, to detect Stuck at zero problem at the input of the first FF in the Following Figure? Assume that initially all DFFs are reset to logic 0.



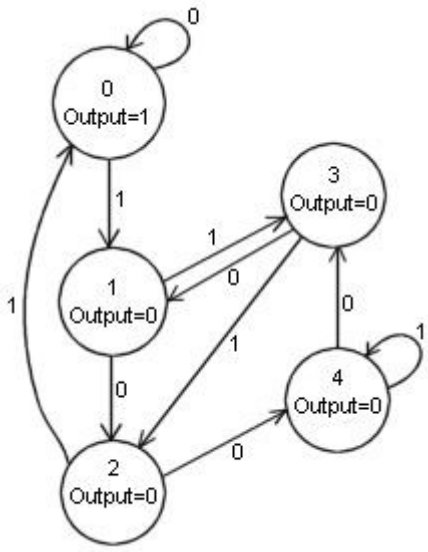
How can I draw a state diagram that asserts a 1 everytime it sees a bit stream that is divisible by 5? For eg: if the bit stream is:

101 output=1 (since 101=5);

1010 output=1 (since 1010=10);

1011 output=0 (since 1011=11) and so on.

We don't need to keep track of the entire string of numbers - if something is divisible by 5, it doesn't matter if it's 250 or 0, so we can just reset to 0. So we really only need to keep track of "0" through "4".



SO=remainder 0

S1=remainder 1

[75]
input A:n bit, B: n bit C:1 bit

if C=1 then out=A-B;
if C=0 then out=A+B;

ANS of Q46: was left unanswered. i feel in this if we could look at the output of Full Adder as combined result.i.e.
if a,b,c are inputs to FA and S,C are outputs then the number of 1's is equal to the concatinated result of S and C in the order CS.

Q78:
If F(A,B,C,D,E) = BE, how many terms will be there in the standard or canonical SOP representation of F?
ANS:
8 terms. BE=(A+!A)(C+!C)(D+!D)BE

1. describe a CMOS
2.Describe the 5 regions of VTC
3.what is the difference between latch and flip-flop
4.given 2 latch's how will you convert into flipflop
5.why do we prefer flipflop to latch
6.draw a 4 bit counter
7.what is setup time and hold time
8.why do we require setup and hold times.
9.what is the difference between a combinational and sequential circuit.

A1. CMOS means Complementary MOS design where i nwe use PMOS and NMOS as pull up and pulldown chains to provide low resistance path from output to VDD or GND. This style of design has good noise margins but requires more number of transistor. We have discussed this question in this thread...
A2.By VTC do you mean the Voltage Transfer Characteristics of Inverter.
The five regions are when Vin = 0, 0-VIL , VIL-VIH, VIH-VDD, VDD
VIL-VIH is the high gain region and digital design avoids working in this region whereas analog design embraces it
A3.Latch is level sensitive where as FF is edge sensitive
A4.We can build a Flip Flop from 2 latches by using Master Slave Configuration where in you give clock to one latch and inverted version of the clock to the other latch... sometihng like ur JK flipflop...
A5.Have 4 FF's .connect the outputs of one FF to the input of the next flip flop...This is a ripple counter..(X) it’s a shift register.
The Qbar of the FF should go on to the clock of the next flipflop, this is for an asynchronous counter

Ans7)Setup Time :Minimum time before the rising/falling edge of the clock for which Data should be stable.

Hold Time: Minimum time after the rising/falling edge of the clock for which Data should be stable.

Ans8) setup time models the time required for the data to propagate within the component and load into the latch before the latch is triggered by the clock.

Hold time models the min time for which the latch looks at its input(after the clock is asserted). After the Hold Time inputs are locked, latch no longer looks at its inputs. Thus Hold time is used to avoid Race conditions.

Q80) How will u interface a 16-bit microprocessor to a 2 1-K RAM.Describe the read and write operations with the necessary signals?Guys this is a basic question asked in one of my interview.

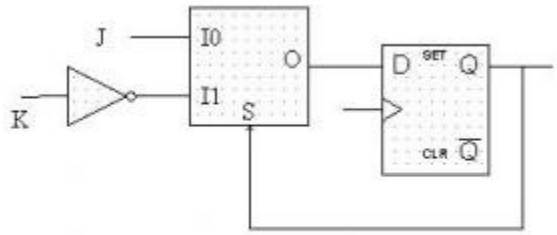
Q81)

Design a circuit for finding the 9's compliment of a BCD number using 4-bit binary adder and some external logic gates.

Ans to Q81)

9's compliment is nothing but subracting the given no from 9.So using a 4 bit binary adder we can just subract the given binary no from 1001(i.e. 9).Here we can use the 2's compliment method addition.Is my answer correct.

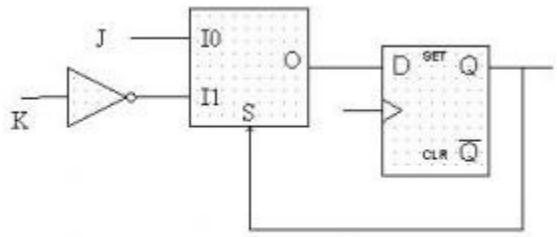
Ans:82:



Q82:

Design a J-K flip flop using DFF,2:1 Mux and an-inverter?

Ans:82:



Q87

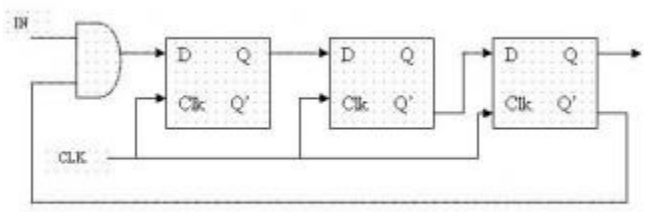
Design a divide by 3 clock. It was asked in one of my friends interview. I guess this has complex logic to it. In a phone interview how do you explain all that stuff, or is there a simple design for it..

Q.87

How do we design a 1Mb SRam? Give the architecture of the same?

Q73:

What are all the test patterns needed at the input, IN, to detect Stuck at zero problem at the input of the first FF in the Following Figure? Assume that initially all DFFs are reset to logic 0.



Give cont. 1s at the input

11111111.....

If there is no stuck at zero, the O/P will be 000000.....and feedback is 111111...

Bcoz as the flops are initially reset to 0, The initial input of AND gate will be 1..if I give 1 at the external input also, the OUT after 3 clocks has to be zero...

But if there is a stuck at zero problem, O/P will be conts 1s instead of 0's....

Q89)

2 bit gray code sequence (00,01,11,10) is coming serially. Design the circuit to detect the false sequence.

Q90.

If u want to convert X xor Y to X or Y ,what should Xand Y be?

1)Antonyms

2)SYNONYMS

3)twos compliment of each other

4)ones compliment

Consider a single bit case,

If X = Y = 1 never occur, XOR is equivalent to OR gate

So now for multiple bit cases, if X and Y are 1's complement of each other, X will never equal to Y. Similarly antonyms (thouh I am not clear about what do u mean by antonyms)

so my answer is (4)

i think if we look at 2's complement even that works.

if a=101 and b=2's complement of a i.e. 011 then the XOR result is 110 which is equivalent to the OR result for the same.

Q91)how to obtain a 3 input OR gate using 2 input NAND gates?how many nand gates are used?

Q91
This is the one of the interiview question sireesha was asked for asic design job
there is a block with 3 i/ps. and 1 o/p
block is +ve edge triggered.
the conditions are
if reset=1,q=0,
else if set=1,q=1,
else q=q.

they are recruiting for ASIC....there were 8 questions....50 marks....first 7 questions were from digital electronics....
1)Match the following:based on clock skew....poles and zeros..IIR FIR filter
2)was simple....implementation of an expression using MUX
3)Had to find the max. freq. of a ckt....with skew,setup and hold time given
4)Draw the state diagram for a divide by 5 ckt....Find the no. of states...Generalize it for divide by N...Ans.there are N-1 states
5)One question where in the areas occupied by multiplier,MUX and registers were given and question was to modify it such that the area in minimised.
6)One algo to store an error report from 3 files into 1 file....only the data from specified fields in each file to be stored in the destination file.
7)Don't remember
8)An apptitude qn....100 doors....1 st time u open doors with nos-multiples of 1...then close doors with multiples of 2...then open---3....repeat till 100...ANS...Doors with nos.1,4,9,16,25,36,49,64,81,100 are open-all square nos.

昨天问过 NV 的 HR，说是不能 take 两个部门的考试试题，而且 90 分钟的时间，确实是有点短了，不过也要看题目的难度了。要求全文用 en 回答，Jolin 也到现场了，看来答完题目要行注目礼了：）

ASIC 流程
解释 setup/hold time violation
PMOS/NMOS 为了平衡 rising/falling time，已知 NMOS 的宽长比，求 PMOS 的宽长比
LATCH/DFF 的最大区别是什么
2 分频电路，4 分频电路，画出实现电路，画出波形
画出边沿出发的主从触发器的 schematic
给了一道判断 setup/hold time 的时序分析的题目
去年的题目再出一次：

net 延时问题，比如 1000um 为 1ns， 10000um 延迟是多少，10 个 1000um 串连,中间以 buffer 隔开，每个 bufer 是 2ns 延迟，问总共延迟为多少?
两道编程的题目，基本的 C 语言和 PERL

感觉笔试题目，注重基础，注重一些电路的概念。所以我们实验室的应聘后端的，应该都不错。

下午面试，Jolin 跟他们经理推荐过了，然后经理看过了我的简历和答题，觉得更适合于到前端去做 architect，于是还是带着准备好的后端 slides，去面试 architect 去了。Radison 酒店，NV 真有钱。不过接待的 mm 到不是很 nice，冷冷的说，“你等会儿吧，好了，你上去吧...哦，好，你可以走了”anyway 幸好不是她面我。先到 24 楼面技术，一个跟我们年级差不多的 engineer 面技术，谈了一下在 Grenoble 做的通讯体系的项目，问了一下细节的问题。然后甩开项目，开始问 C/C++，

访问 memory 怎么避免 memory 的 leakage，
怎么用正则表达式表示实数;
以及 Graphic 算法方面的问题，怎么知道 n!后面会有多少个“0”，例如 5!的结果末尾会有 1 个零;
一个立方体，怎么知道表面上两点之间的最短距离（沿表面的最短距离）
气氛还是瞒好的，因为大家都是同龄人，没有什么距离感。回到 1657,等 Architect manager 面完上一个人，结果从 24 楼下来，在 16 楼打个照面，再上到 24 楼，就在刚刚面人的隔壁。在宣讲的时候，已经跟 manager 聊过一段时间，这次过来 manager 也很 nice 的先再详细介绍了一下他手下三个组各自的职责，然后介绍了一下 Graphic 方面做 architect 跟做 CPU 的不同，主要还是先让我考虑清楚了做 GPU 跟 CPU 的不同。同时介绍了一下为什么我 take 了 physical design 的试卷，结果来面 architect。是 physical design 的 manager 亲自电话过来，说看过我的简历试卷以后，觉得这样的背景应该更适合 architect（多谢 Jolin 的推荐了），因为后来博士论文也在做这方面的工作，所以也有这样的兴趣来做 architect 方面的工作。介绍了一下做过的体系的工作，然后开始问一些问题:

一个 8bit*8bit+8bit 的计算，问最多用到多少位的 reg;
request1: C 语言去访问 memory，给出 size，分配一段地址，然后返回地址；request2: 只有 request1 返回的地址是 16 的倍数时，才返回正确信息；问怎么实现。

前面那道题倒是想了想，回答出来了，后面这个就只是了解了下题目意思，后来也没有时间了，也没有在启发下得到最后结果。毕竟还是做的不多。不过如果能去 architect 组，首先要调整自己硬件思维的方式，要用体系的方式来想问题，然后再是用 GPU 的方式来思考问题。

总的来说，不算太好，但是也强调了自己的优势是什么，因为项目这种事情，毕竟不是自己决定的，自己下面了解的东西，没有实际的项目经验，还是要欠缺的，不过已经尽力了，该表现自己强项的地方，都表达出来了，只有等两个星期看结果了。

1.画二分频电路
,占空比 3:1。2. 画 FSM 状态图,输出方程 3. 7 输入 3 输出,用基本门搭建电路，统计输入中 1 的个数。4. synchronous reset vs asynchronous reset?pros and cons?5.DFF 的反馈回路，计算最大时钟 6。MUX+ADDER 结构，优化至 300MHZ 7.hold time violation,降低时钟能否解决 8.低功耗优化。

全加器的结构，门级示意图，布尔表达式，超前进位加法器原理。
ASIC design flow 的主要流程
Fifo， 堆栈都是什么，主要功能

Verilog 语言的简单 coding，以及阅读，画波形

简单时序电路的 timing 分析和优化， 关键是 setup/hold time，不懂没关系，半个小时可以学会。

计算机主要组成部分，体系结构最简单的知识，cache。（如果不知道南桥，北桥是做什么的，就有点过分了）

最基本的电路分析，电阻，电容，电感。（串联电容的分压要知道怎么算）

简单的电容冲放电计算

Cmos 基本知识。（二输入与非门总要会画吧）

Unix 环境下几种常见的脚本语言，tcl，shell，perl，懂一点就行。

以上这些内容都是我的笔试题，而且不同公司的笔试题还有重复的部分，比如加法器，比如 setup/hold 等等。